

Monte Carlo Simulation for Option Pricing

Monte Carlo simulation is a numerical technique used to estimate the value of financial derivatives by simulating a large number of possible future price paths for the underlying asset. Unlike the Black–Scholes model, which provides a closed-form analytical solution under specific assumptions, Monte Carlo methods rely on repeated random sampling to approximate the expected payoff of an option.

In option pricing, the underlying asset price is typically assumed to follow a **Geometric Brownian Motion (GBM)**, and its future price at maturity is modeled as:

$$S_T = S_0 \exp \left[\left(r - \frac{\sigma^2}{2} \right) T + \sigma \sqrt{T} Z \right]$$

where:

- S_0 = Current asset price
- S_T = Asset price at maturity
- r = Risk-free interest rate
- σ = Volatility of the underlying asset
- T = Time to maturity (in years)
- Z = Standard normal random variable ($Z \sim N(0,1)$) ($Z \sim N(0,1)$)

For each simulated price path, the option payoff is calculated. For a European call option:

$$\text{Payoff} = \max(S_T - K, 0)$$

where K is the strike price.

The option price is then estimated as the discounted average payoff across all simulated paths:

$$C = e^{-rT} \frac{1}{N} \sum_{i=1}^N \max(S_T^{(i)} - K, 0)$$

Where

N = Number of simulations

$S_T^{(i)}$ = it is the i th simulation simulated price near maturity

As the number of simulations increases, the estimated option price converges to the theoretical expected value.

Advantages

- Suitable for pricing complex and path dependent derivatives.
- Flexible enough to incorporate different stochastic models and market assumptions.
- Easy to extend for multi-asset and exotic options.

Limitations

- Computationally intensive, especially when high accuracy is required.
- Convergence is slower than analytical methods such as the Black–Scholes model.
- Results depend on the quality and number of random simulations.

